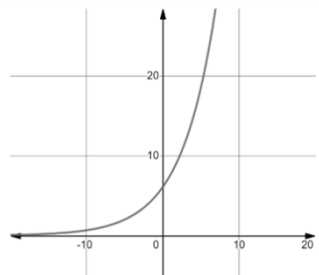


The table shows three different functions.

Function A	Function B	Function C												
$y = 2x + 2$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4	
x	y													
-2	4													
-1	1													
0	0													
1	1													
2	4													

1. Determine which type of function that each represents.

Function A	Function B	Function C
☐ Exponential ☒ Linear ☐ Quadratic	☐ Exponential ☐ Linear ☒ Quadratic	☒ Exponential ☐ Linear ☐ Quadratic

2. Which function has the largest y -intercept?

Function C

3. Compare the ranges of the functions and indicate any similarities and differences.

The range of Function A is all real numbers. Functions B and C have similar ranges, but not the same. Function B's range is $y \geq 0$, and Function C's range is $y > 0$.

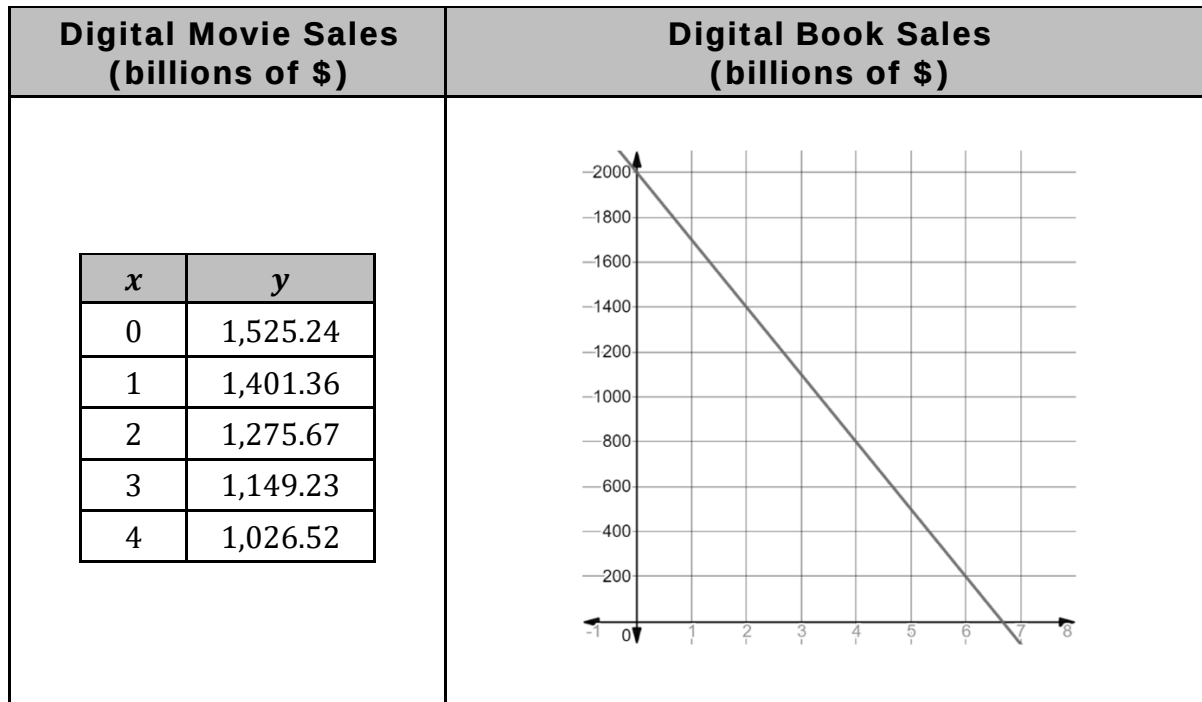
4. Compare the domains of the functions and indicate any similarities and differences.

All three functions have the same domain – all real numbers.

5. As $x \rightarrow \infty$, which function has larger values of y ?

Function C

The table shows a table of values representing the revenue, in billions of dollars, from sales of digital movies, where $x = 0$ is 2015, and a graph representing the revenue, in billions of dollars, from sales of digital books, where $x = 0$ is 2015.



6. Approximate the x -intercept for the function that represents the sale of digital books.
(6.6, 0)
7. Interpret the x -intercept of the function that represents the sale of digital books.
Approximately 6.6 years after 2015, there will be no revenue from sales of digital books.
8. What is the domain of the function representing the sale of digital books?
 $\{x|x \in \mathbb{R}\}$
9. Which function has a larger y -intercept?
Digital book sales
10. What type of function best describes the table of values that represents digital movie sales? Explain your choice.

- Exponential
 - Linear
 - Quadratic

The rate of change is fairly consistent from year to year.